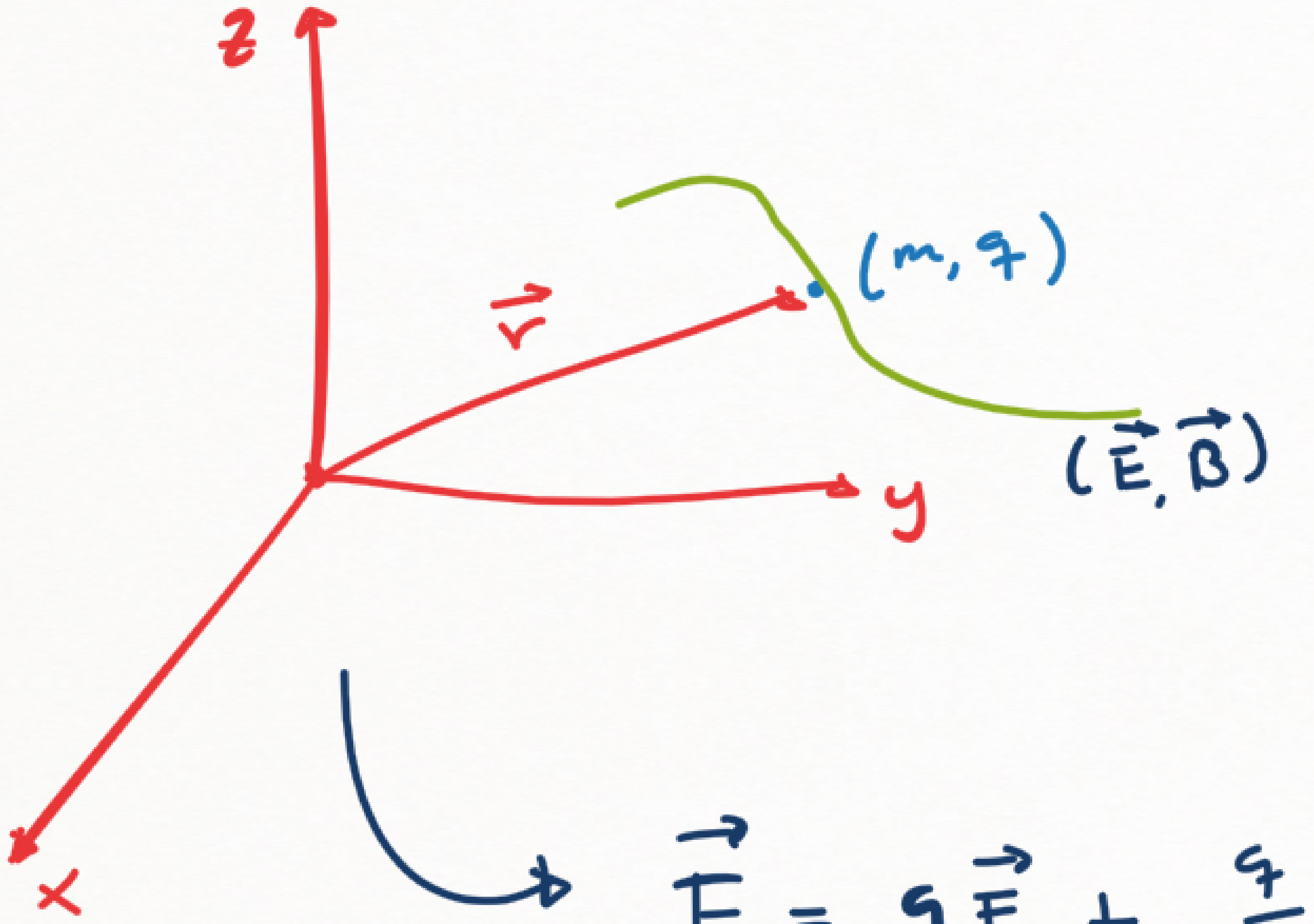
Mecánica teórica — Clase # 4 — 12/08/2025



$$\vec{F} = q\vec{E} + \frac{q}{c}(\vec{v} \times \vec{B})$$

$$(\vec{E}, \vec{B}) \longleftrightarrow (\phi(\vec{r}, t), \vec{A}(r, t))$$

$$\vec{E} = -\nabla\phi - \frac{1}{c}\frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A} = \varepsilon_{ijk}\partial_j A_k$$

$$\vec{F} = -\nabla U + \frac{d}{dt}(\nabla_{\vec{v}} U)$$

$$Q_i = \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_i}\right) - \frac{\partial U}{\partial q_i}$$

$$U = q\phi - \frac{q}{c}(\vec{A} \cdot \vec{v}) = q\phi - \frac{q}{c}(A_x\dot{x} + A_y\dot{y} + A_z\dot{z})$$

$$F_x = qE_x + \frac{q}{c}(\vec{V}\times\vec{B})_x$$

Tensor de Levi-Civita (checar permutación de índices)

Nota:

$$(\vec{A}\times\vec{B})_i = \varepsilon_{ijk} A_j B_k$$

$$(\vec{V}\times\vec{B})_x = \varepsilon_{xjk} V_j B_k = \varepsilon_{xyk} V_y B_k + \varepsilon_{xzk} V_z B_k$$

$$\vdots$$

$$= \varepsilon_{xyz} V_y B_z + \varepsilon_{xzy} V_z B_y$$

$$\vdots$$

$$= V_y B_z - \ldots \quad \ldots$$

Tenemos

$$F_x = qE_x + \frac{q}{c}\left(\vec{V}\times\vec{B}\right)_x = qE_x + \frac{q}{c}\left(V_y B_z - V_z B_y\right)$$

$$E_x = -\frac{\partial \phi}{\partial x} - \frac{1}{c}\frac{\partial A_x}{\partial t}$$

$$B_i = \left(\nabla\times\vec{A}\right)_i = \varepsilon_{ijk}\frac{\partial}{\partial x_j}A_k = \varepsilon_{ixk}\frac{\partial}{\partial x}A_k + \varepsilon_{iyk}\frac{\partial A_k}{\partial y} + \varepsilon_{izk}\frac{\partial A_k}{\partial z}$$

$$B_z = \varepsilon_{zxy}\frac{\partial A_y}{\partial x} + \varepsilon_{zyx}\frac{\partial A_x}{\partial y} = \frac{\partial A_x}{\partial y} - \frac{\partial A_x}{\partial y} - \varepsilon_{zxy}$$

Rotacional mediante la forma de determinante

$$\nabla \times \vec{A} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ A_i & A_y & A_z \end{vmatrix} = \left(\partial_y A_z - \partial_z A_y\right)\hat{i} - \left(\partial_x A_z - \partial_z A_i\right)\hat{j} +$$

$$\left(\partial_x A_y - \partial_y A_i\right)\hat{k}$$

$$\vdots$$

$$= \varepsilon_{ijk}\,\partial_j A_k$$

Tenemos

$$E_x = -\frac{\partial \phi}{\partial x} - \frac{1}{c}\frac{\partial A_x}{\partial t}$$

$$E_y = -\frac{\partial \phi}{\partial y} - \frac{1}{c}\frac{\partial A_y}{\partial t}$$

$$E_z = -\frac{\partial \phi}{\partial z} - \frac{1}{c}\frac{\partial A_z}{\partial t}$$

$$B_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}$$

$$B_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}$$

$$B_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}$$

Tenemos

$$F_x = q E_x + \frac{q}{c}(\vec{V} \times B)_x = q E_x + \frac{q}{c}(V_y B_z - V_z B_y)$$

$$F_x = q\left[-\frac{\partial \phi}{\partial x} - \frac{1}{c}\frac{\partial A_x}{\partial t}\right] + \frac{q}{c}\left[\dot{y}\left(\frac{\partial A_y}{A_x} - \frac{\partial A_x}{\partial y}\right) - \dot{z}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\right]$$

Tenemos

$$F_x = -\frac{\partial U}{\partial x} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{x}}\right) \quad ; \quad \frac{\partial U}{\partial x} = q\frac{\partial \phi}{\partial x} - \frac{q}{c}\left[\frac{\partial A_x}{\partial x}\dot{x} + \frac{\partial A_y}{\partial x}\dot{y} + \frac{\partial A_z}{\partial x}\dot{z}\right]$$

Ahora respecto a $\dot{x}$

$$\frac{\partial U}{\partial \dot{x}} = - \frac{q}{c} A_x \Longrightarrow \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{x}}\right) = - \frac{q}{c}\left[\frac{\partial A_x}{\partial x}\dot{x} + \frac{\partial A_x}{\partial y}\dot{y} + \frac{\partial A_x}{\partial z}\dot{z} + \frac{\partial A_x}{\partial t}\right]$$

Tenemos

$$\Rightarrow F_x = - q\frac{\partial \phi}{\partial x} + \frac{q}{c}\left[\frac{\partial A_x}{\partial x}\dot{x} + \frac{\partial A_y}{\partial x}\dot{y} + \frac{\partial A_z}{\partial x}\dot{z}\right] - \frac{q}{c}\left[\frac{\partial A_x}{\partial x}\dot{x} + \frac{\partial A_x}{\partial y}\dot{y} + \frac{A_x}{\partial z}\dot{z} + \frac{\partial A_x}{\partial t}\right]$$

$$F_x = q\left[-\frac{\partial \phi}{\partial x} - \frac{1}{c}\frac{\partial A_x}{\partial t}\right] + \frac{q}{c}\left[\dot{y}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) - \dot{z}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\right]$$

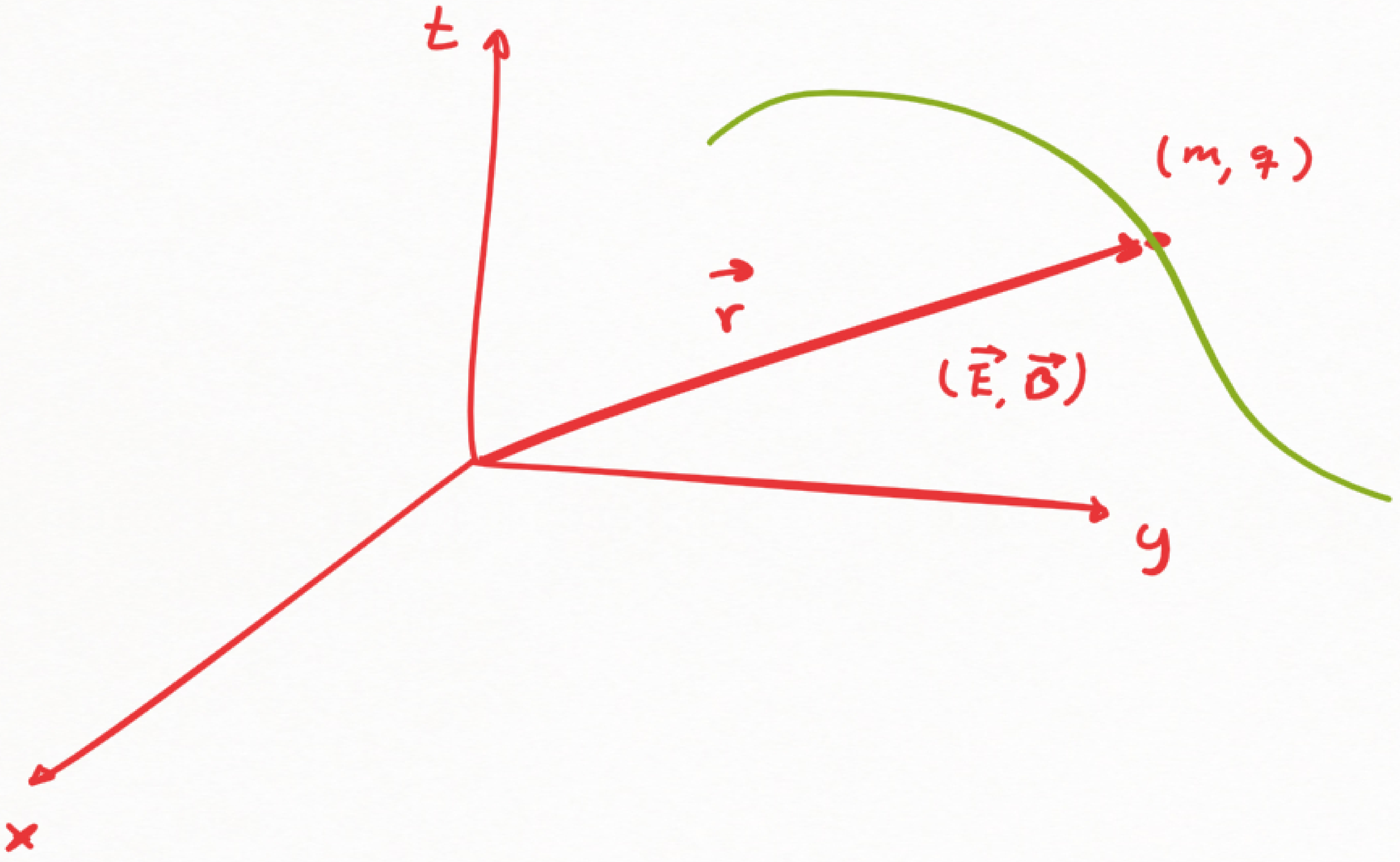
Por tanto la componente x coincide en ambos casos:

↳ Resolver para "y" y "z": (las componentes restantes).

Escribir la fuerza de Lorentz para campo magnético y eléctrico es equivalente a la expresión:

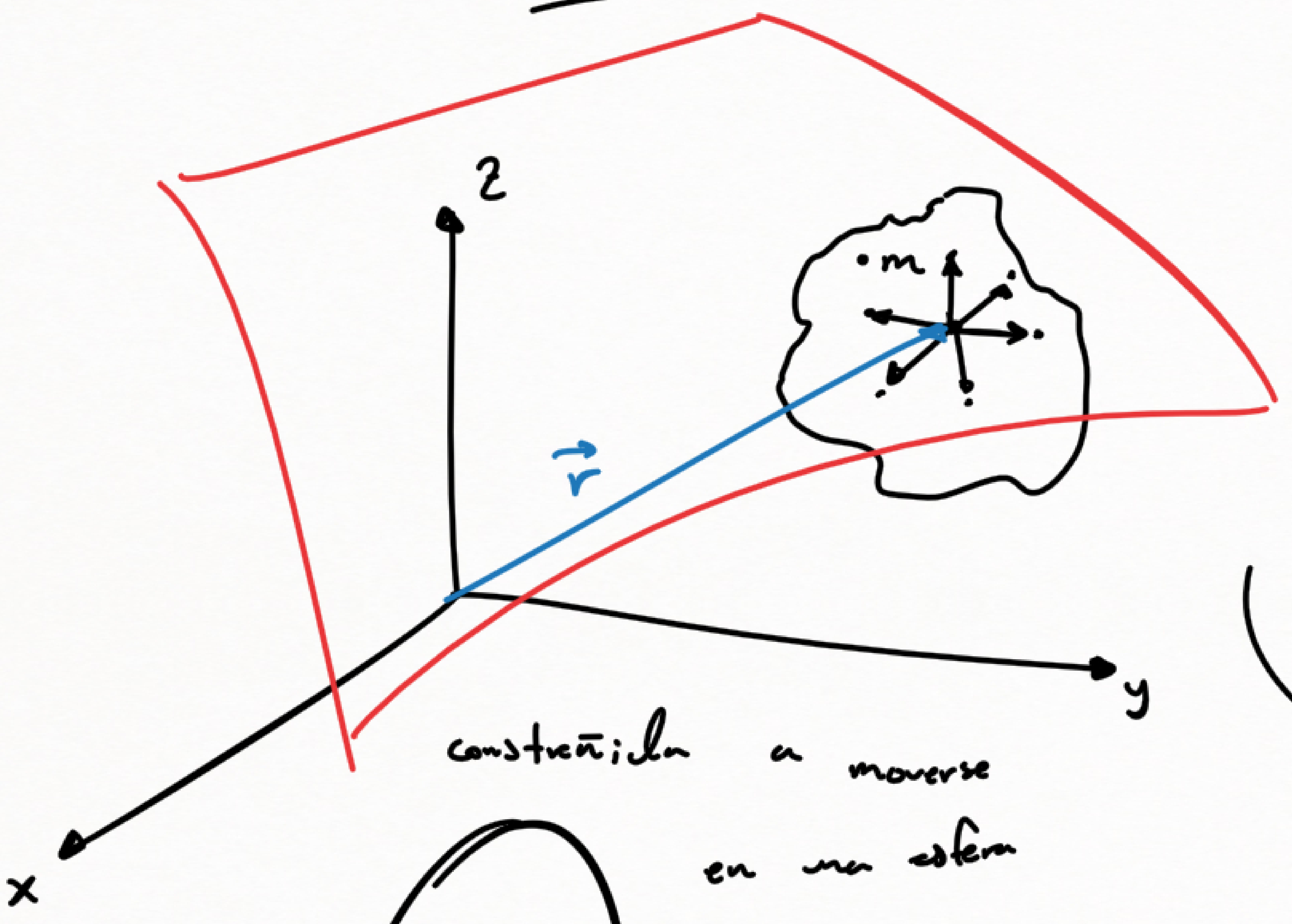
$$\overbrace{\vec{F} = -\nabla U + \frac{d}{dt}\left(\nabla_{\vec{v}} U\right)}$$

Ahora la Lagrangiana es:



- 2 próximas semanas examen Parcial.

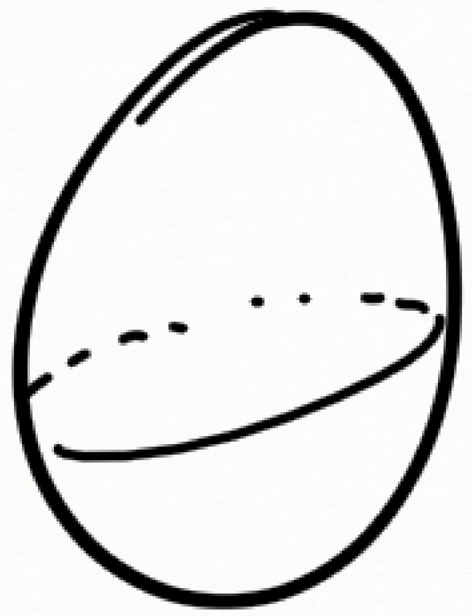
Grados de libertad



$$f(x, y, z, t) = cte.$$

$$z = z(x, y, t)$$

constreñida a moverse en una esfera



Condición de constricción

$$f(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$$

$$g(x, y, z) = x^2 + y^2 + z^2 = cte$$

coordenadas cartesianas rectangulares para la constricción (aunque no sean las mejores)

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = x\hat{x} + y\hat{y} + z(x, y, t)\hat{z}$$

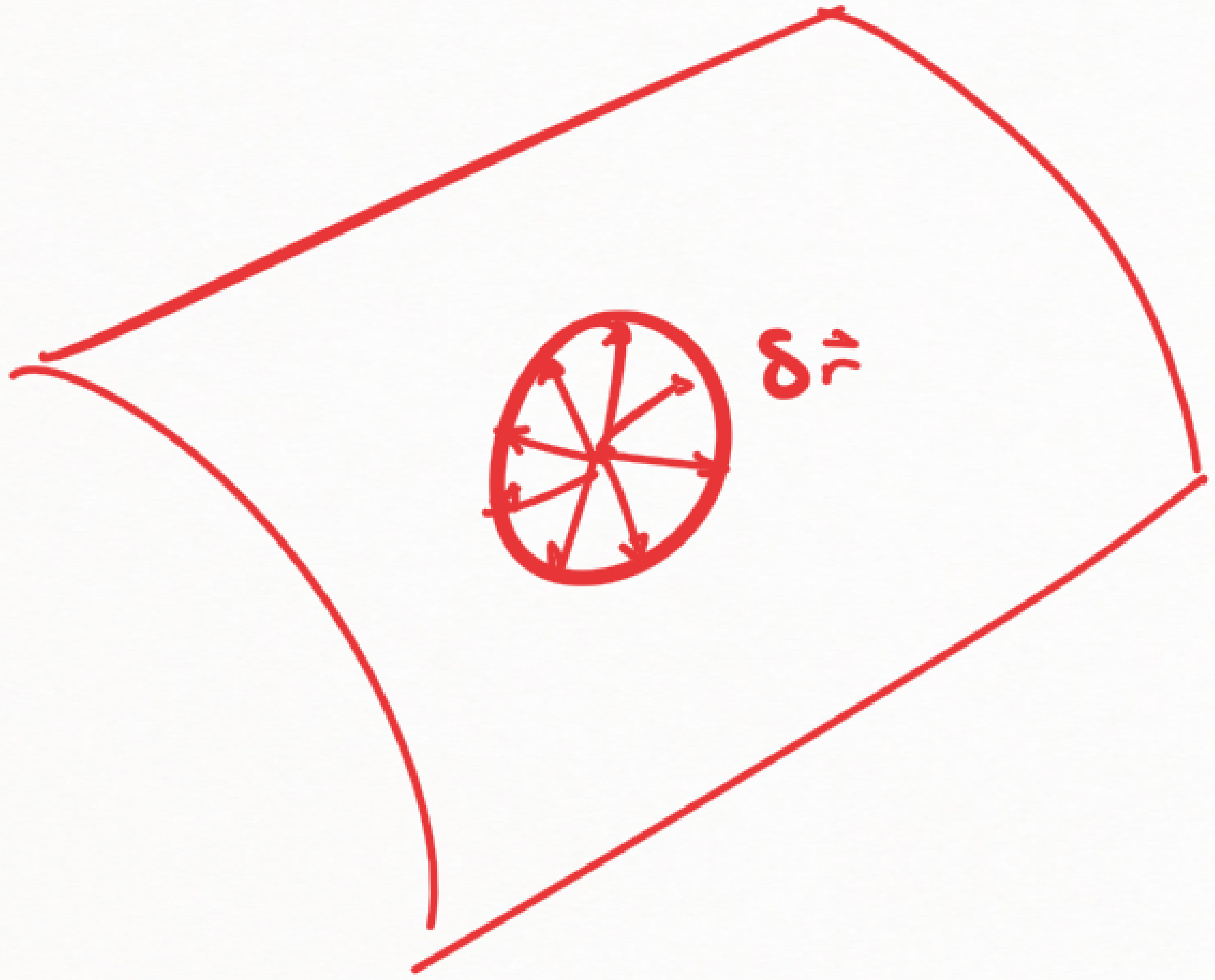
Por segunda ley de Newton:

vector de posición en coordenadas rectangulares

$$m\frac{d^2\vec{r}}{dt^2} = \vec{F} + \vec{F}_c \quad ; \quad \vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{F}_c = F_{cx}\hat{x} + F_{cy}\hat{y} + F_{cz}\hat{z}$$

- Ejemplo donde el principio de D'Alembert no se cumple.



- Hacemos que la fuerza de constricción valga 0

- Las fuerzas de constricción las conocemos luego el vector el sistema.

$$\left(m\frac{d^2\vec r}{dt^2}-\vec F\right)\cdot\delta\vec r = \underbrace{\vec F_c\cdot\delta\vec r}_{\delta w} = 0 \Longrightarrow \boxed{\left(m\frac{d^2\vec r}{dt^2}-\vec F\right)\cdot\delta\vec r = 0}$$